

Magnetic Interference and Calibration: A Comparative Analysis of Magnetometers

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Abstract—Magnetometer Calibration for accurate heading estimation is crucial for robotics and autonomous navigation. In this paper, we obtain 14 datasets from an industrial grade VN-100 and a consumer grade iPhone under controlled hard iron, soft iron, and combined distortion conditions at 7-cm and 30-cm separation distances. Next, we compare four distinct calibration algorithms: MinMax, Magcal, Scalar Extended Kalman Filter (EKF), and Levenberg-Marquardt. Algorithms were evaluated on magnitude error, post-calibration consistency, and computational cost. Contrary to our expectations, the iPhone 11 outperformed VN-100, achieving mean magnitude errors of 13–19% compared to 22–34% for the VN-100. This discrepancy is attributed to the Vector Processing Engine introducing inter-sample dependencies and violating the independence assumption. Among the algorithms, Magcal and Levenberg-Marquardt performed the most consistent with mean standard deviations below $1 \mu\text{T}$ on iPhone data. These results show that data sensor processing can be far more important than sensor quality in calibration performance.

I. INTRODUCTION

Magnetometers are sensors that measure magnetic field strength and direction. These sensors are used for navigation in robots, drones, and consumer devices for heading and direction. However, magnetometer reading accuracy is highly vulnerable to two types of distortion: hard-iron caused by permanent magnets near the sensor, which introduce a constant offset to all measurements, Soft-iron caused by ferromagnetic materials that warp the shape of the local magnetic field, producing orientation-dependent errors.

Calibration aims to correct for these distortions by estimating and removing the soft and hard iron components from the raw sensor data. Many algorithms exist from more simple geometric approaches to iterative non-linear optimization, but their relative performance have not been compared. More specifically, it is not clear whether algorithm choice or sensor characteristic is the most important factor in calibration accuracy.

In this project we aim to compare 4 algorithms, MinMax, Magcal, Scalar Extended Kalman Filter, and Levenberg-Marquardt, across two sensors: an industrial grade IMU, the VN100, and a consumer grade IMU, an iPhone 11. Data was collected under controlled conditions with intentional soft, hard, and soft/hard distortions at two distances, 7 cm and 30 cm. The algorithms are evaluated on magnitude error, post-calibration consistency, and computational cost. Our key finding was that the consumer grade iPhone 11 produced a more accurate and consistent calibration result than the VN-100. This counter-intuitive result is attributed to VN-100's internal Vector Processing Engine violating the independence assumptions made by the algorithms.

II. RELATED WORK

Magnetometers are widely used in robotics, mobile devices, and inertial navigation systems because they provide an absolute heading reference with respect to the Earth's magnetic field. However, magnetometer measurements are highly sensitive to local magnetic disturbances, making calibration a necessary step before reliable heading estimation can be performed. A common family of magnetometer calibration

methods is based on ellipsoid fitting. Renaudin et al. [7] presented a complete triaxis magnetometer calibration method in the magnetic-field domain, modeling scale factors, offsets, hard-iron, and soft-iron effects. Their work is closely related to the ellipsoid-based calibration methods evaluated in this project because both approaches treat calibration as the problem of recovering an undistorted magnetic field norm from distorted three-axis measurements. Vasconcelos et al. [8] also formulated magnetometer calibration geometrically, estimating calibration parameters from vector observations and emphasizing the importance of correcting both additive and multiplicative distortions. These methods provide the theoretical basis for algorithms such as Matlab's magcal, which estimates a correction matrix and bias vector from a set of magnetometer samples.

Other calibration approaches use nonlinear optimization to refine calibration parameters. Levenberg–Marquardt methods are commonly used when the calibration objective is expressed as a nonlinear least-squares problem, such as minimizing the difference between the calibrated magnetic-field magnitude and the expected local field magnitude. Li and Griffiths [9] described least-squares ellipsoid-specific fitting, which is relevant to magnetometer calibration because distorted magnetometer measurements often lie on an ellipsoid before correction. Liu et al. [10] proposed a two-step optimized magnetic correction algorithm using Levenberg–Marquardt fitting with ellipsoid constraints to compensate for hard-iron and soft-iron effects. Compared with simple range-based methods such as minmax calibration, nonlinear and ellipsoid-fitting methods use the full measurement set and are generally more robust when the sensor has been rotated through many orientations. Kalman filter based approaches estimate calibration parameters recursively as new measurements arrive, making them attractive for online calibration in mobile robotics and navigation. Crassidis et al. [11] provide a broad survey of nonlinear attitude-estimation methods, including extended Kalman filtering, which is relevant because magnetometer calibration and attitude estimation both involve nonlinear measurement models.

In robotics and inertial navigation, calibration quality is also affected by the sensor platform and by any preprocessing performed before data are exported. Industrial IMUs often include onboard filtering, attitude estimation, and magnetic-disturbance rejection. The VN-100, which has been the basis of our measurements in the course and in part of this project as well, includes a Vector Processing Engine with real-time magnetic and acceleration disturbance rejection, adaptive signal filtering, dynamic filter tuning, and onboard hard/soft-iron compensation. This is important for our project because several calibration algorithms assume that the input samples are raw, independent observations of the magnetic field. If the sensor internally filters or compensates measurements before export, the resulting data may violate these assumptions and affect the behavior of offline calibration algorithms, which is part of the interest in our investigation.

III. METHODOLOGY

A. Algorithms

For this study, we implemented and compared four different algorithms: MinMax, Magcal, Scalar Extended Kalman Filter, and Levenberg-Marquardt.

1) *MinMax*: The MinMax algorithm calibrates Magnetometer data by subtracting the hard iron bias from the raw magnetometer data and then multiplying that result by the soft iron scale factor. See the code below for the calculations applied to each axis, X, Y, Z.

$$\text{cal} = (\text{raw} - \text{hard}) \cdot \text{soft} \quad (1)$$

$$\text{hard} = \frac{\text{axisMax} + \text{axisMin}}{2} \quad (2)$$

$$\text{soft} = \frac{\text{avgRange}}{\text{axisRange}} \quad (3)$$

This algorithm is unique in that it only looks at six extreme points within the dataset, making the algorithm fast but sensitive to outliers.

2) *Magcal*: The Magcal implementation is a built-in function in MATLAB[3]. Magcal employs an ellipsoid fitting algorithm to calibrate the magnetometer data. See the code below used to implement this algorithm.

```
magXYZ = [magX, magY, magZ];
[A, b, expMFS] = magcal(magXYZ);
magXYZCal = (magXYZ - b) * A;
```

Magcal computes a matrix, A, and a vector, b, which are applied to the uncalibrated magnetometer data in the same manner as the MinMax algorithm. Magcal also computes the expected magnetic field strength, expMFS. Magcal differs from MinMax in that Magcal looks at every point in the dataset.

3) *Scalar Extended Kalman Filter*: The Scalar Extended Kalman Filter is initialized with a linear least squares sphere fit in order to estimate the initial bias vector and expected field magnitude. The algorithm then processes each data point. At each step the algorithm compares the measured magnetic field with the expected magnetic field and updates the bias estimate and uncertainty as it goes. This algorithm is scalar because it collapses the 3D measurement vector into a single number. It is extended because the filter linearizes a nonlinear magnetic field magnitude function using its Jacobian.

This algorithm is unique in that it looks at the data sequentially. This means that the algorithm can adapt if there is a change in the bias during data collection. This algorithm also only accounts for hard iron distortion. It is possible to modify this algorithm to handle soft iron distortions using the ellipsoid fitting step in the Levenberg-Marquardt algorithm, but this was not implemented in order to keep both algorithms independent for analysis.

4) *Levenberg-Marquardt*: Like the Scalar EKF algorithm, the Levenberg-Marquardt algorithm initializes with a linear least squares sphere fit to get an initial hard iron and radius estimate. Then, the algorithm iteratively minimizes residuals between the measured field magnitude and the fitted radius. The algorithm then refines the soft iron correction through an

ellipsoid fit on the hard-iron-corrected data. This algorithm is most similar to Magcal, but Levenberg-Marquardt solves the hard iron problem non-linearly and iteratively rather than with linear algebra all at once. This generally produces a more accurate hard iron estimate, especially when dealing with data that has been exposed to severe distortions.

B. Code Availability

The full implementation of all calibration algorithms and analysis scripts is available on GitHub. While algorithm implementation is the same across both GitHubs, both repositories generate different plots.

For MATLAB algorithms and plots see: <https://github.com/howe-mal/Magnetometer-Comparison-Project>

For Python Algorithms and plots see: https://github.com/anramz29/Magnetometer_Comparison_Project.

IV. EXPERIMENTAL SET UP

For the collection of real-world data three types of material were required. Sensors were used for collecting and reading the data. Hard and soft iron distorters were used to create measurable disturbances in the data. And boxes were used to create distance between the sensors and distorters while keeping them in the same frame of reference.

A. Hardware

1) *VN-100*: For this experiment, data was collected using an industrial-grade IMU, a VN-100, and a consumer-grade IMU, an iPhone 11. The VN-100 uses a 3-axis MEMS magnetometer with minimal sources of internal magnetic interference. A VN-100 sensor, similar to the one we used in this experiment, costs around 1,000 dollars[5]. Data was collected from the VN-100 via wired access. The data was collected into ROS bags containing db3 files as well as the associated metadata. The VN-100 allowed us to collect data at a sampling rate of 40 Hz.

2) *iPhone 11 With Sensor Logger*: The iPhone 11, on the other hand, uses a MEMS Hall-Effect Sensor with large sources of internal magnetic interference from items such as the speakers and magsafe components. While a refurbished iPhone 11 costs around 200 dollars[1], the app used to record and export the data, Sensor Logger[2], was free. Data was collected wirelessly through the phone to a CSV file which was then exported via AirDrop to a computer. The iPhone 11 capped the sampling rate at 25 Hz.

B. Distortion Materials

For this experiment we also used items to stimulate hard and soft iron distortion. For hard iron distortion, a powered-off Apple Watch A1858 was used. For soft iron distortion, two steel bolts were used. To test the effect of the distortions at different distances, two separators were used. To create seven centimeters of distance between the sensor and the distorter, a crayon box was used. To create thirty centimeters of distance, a shoe box was used. This set up can be seen in Fig. 1. and Fig. 2. During the experiments the sensor was taped to one

side of the separator while the source of distortion was taped to the other side. This ensured that the sensor and source of distortion were in the same frame of reference throughout the data collection.



Fig. 1. Seven centimeter setup for soft iron distortion collected via VN-100.

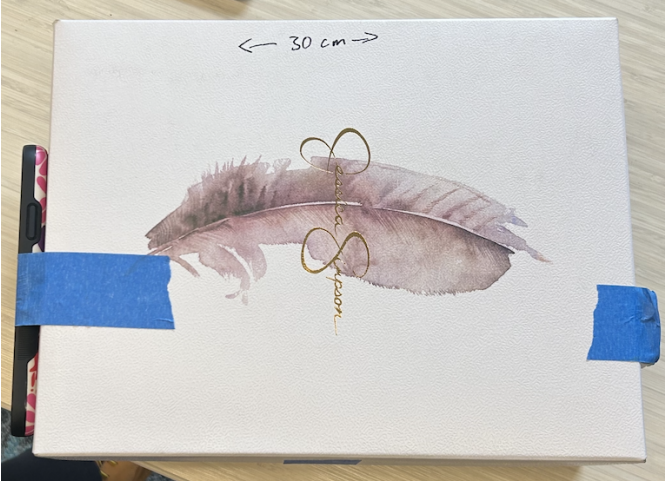


Fig. 2. Thirty centimeter setup for hard iron distortion collected via iPhone.

C. Data Collection Procedure

For data collection, we made sure to clear the environment of as much external hard and soft iron distortions as possible, standing approximately the same distance from the laptops collecting the data for each collection. This ensured that if there were any outside distortions they were consistent across all datasets. For this study, we collected 14 datasets.

This data was collected by rotating the sensor to as many angles as possible, drawing figure-eight circles in the air for approximately two to three minutes. First, clean data was collected. Then, data was collected with intentional hard iron distortion only. The same was done with intentional soft iron distortion only. Finally, data was collected with simultaneous intentional hard and soft distortion. These three scenarios were repeated twice, once with seven centimeters of separation and once with thirty centimeters of separation. This whole process was done twice, once with the VN-100 sensor and then again with the iPhone 11.

D. Evaluation Metrics

After running the calibration algorithms, the results were analyzed by comparing the calculated field magnitude with the expected field magnitude of Boston, approximately $52 \mu\text{T}$.¹ The standard deviation, measured in μT , of the magnitude of the calculated field magnitude was calculated. The percentage error was calculated as the difference between the mean calculated field magnitude and the expected field magnitude, divided by the expected field magnitude and multiplied by 100 to yield a percentage. Computational cost was measured during the calibration process as algorithm run time. This was done using MATLAB's `timeit` feature[4]. See implementation below.

```
tKalman = timeit(@()...
    runKalmanFilter(magX, magY, magZ));
```

V. RESULTS

A. Sensor Comparison

The most surprising result in this study was discovering that the calibration on the data collected with the consumer-grade IMU, the iPhone 11, was more accurate and more consistent than the calibration on the data collected with the industrial-grade IMU, the VN-100. The consistency of this difference, see Fig. 4, regardless of set up or algorithm indicates two possibilities for this discrepancy.

The hard iron bias for the iPhone data was consistently estimated to be between 450 and $500 \mu\text{T}$. This large hard iron bias is likely due to the internal magnetic components of the iPhone discussed earlier. On the other hand, the hard iron bias for the VN-100 was consistently estimated to be less than $50 \mu\text{T}$. This could suggest that the algorithms were overcorrecting for a bias that was not largely present in the VN-100 data, leading to a higher magnitude percent error and standard deviation.

It is also important to note that the VN-100 contains a Vector Processing Engine. This component applies real-time filtering [6] to data collection to mitigate issues such as magnetic and acceleration disturbance as the data is being collected. These processes create dependencies between data points. This is a likely source of the poor performance as the calibration algorithms assume that all data points are independent. We

¹The expected field magnitude is an estimate of the location in which the data was collected (Boston) we do not know the exact field magnitude.

Raw vs Calibrated: Datasets 3, 4, 5

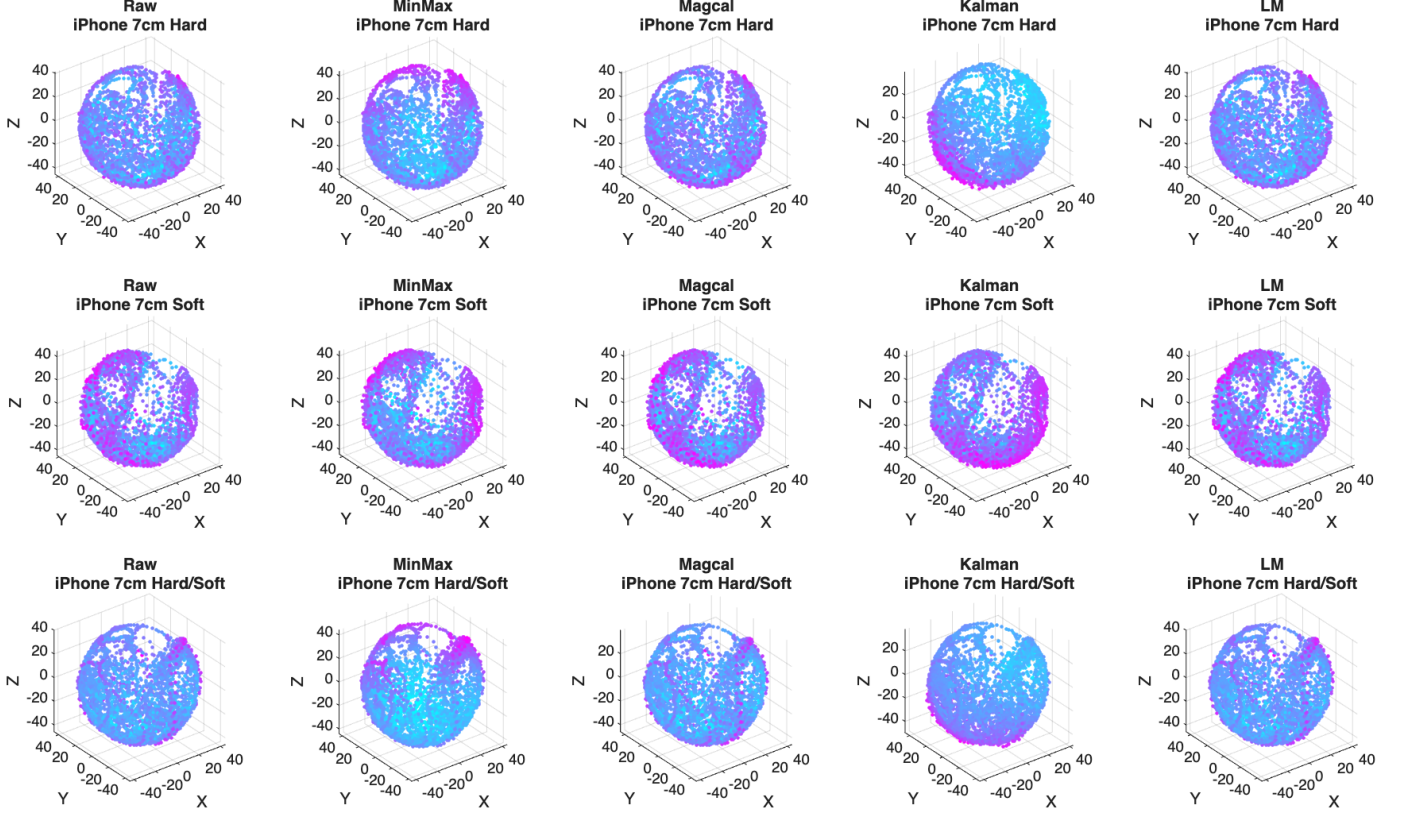


Fig. 3. 3D scatter plots of raw and calibrated magnetometer data for iPhone 11 datasets under 7cm distortion conditions. Color indicates distance from the origin, uniform color indicates consistent field magnitude after calibration.

do not see similar effects in the iPhone data as the iPhone 11 and the Sensor Logger App do not apply any real-time filtering, the data is exported raw and unprocessed.

B. Algorithm Comparison

When looking at runtime in Fig. 5, MinMax is by far the best algorithm, remaining consistently fastest across all datasets. This is due to the fact that only 6 extreme data points are analyzed rather than the full dataset. Scalar EKF is consistently the slowest algorithm. This is likely due to the sequential manner in which the data is processed, with a predict and update step for each sample. Magcal and Levenberg-Marquardt have comparable runtimes, with Magcal performing slightly better than Levenberg-Marquardt most of the time.

The effectiveness of each algorithm to correct hard and soft iron distortions can be visualized in Fig. 3. In this figure, each row represents a different condition, soft iron distortion only, hard iron distortion only, or hard and soft iron distortion. The raw data collected in each condition is seen in the left column. The calibrated spheres are shown in the remaining four columns, each column corresponding to an algorithm.

Looking at the first row, these plots show that, under hard iron distortion, all algorithms were able to center the sphere. MinMax shows the most color variation, suggesting the algorithm is sensitive to hard iron distortions. As expected,

Magcal and Levenberg-Marquardt yield very similar spheres and coloring. The Scalar EKF sphere in this row stands out, with large clusterings of very pink points on one end and a gradient towards light blue points on the other. This is most likely a limitation of the sequential method. The bias estimate evolves over time, meaning early and late measurements receive different corrections depending on when in the collection sequence each orientation was sampled.

The soft iron distortion plots show that the data is a lot more difficult to calibrate. These spheres are noticeably less uniform, with significantly more variation in color. This illustrates how soft iron is the most difficult condition for all the algorithms to handle.

Under both hard and soft iron distortions, all the spheres look relatively similar, as blue with moderate variation. MinMax stands out with the most variation in color and a few clusters of pink points, indicating that algorithm struggled the most in this setting.

Another important metric to look at is performance under different conditions and distances. This examined using the calculated magnitude error percent and calculating the degradation in algorithm performance from thirty centimeters down to seven centimeters, as seen in Table I. The expected behavior for this metric is a positive degradation, as the hard and soft iron distortions move closer to the sensor, the distortion becomes more severe and should be harder for the calibration

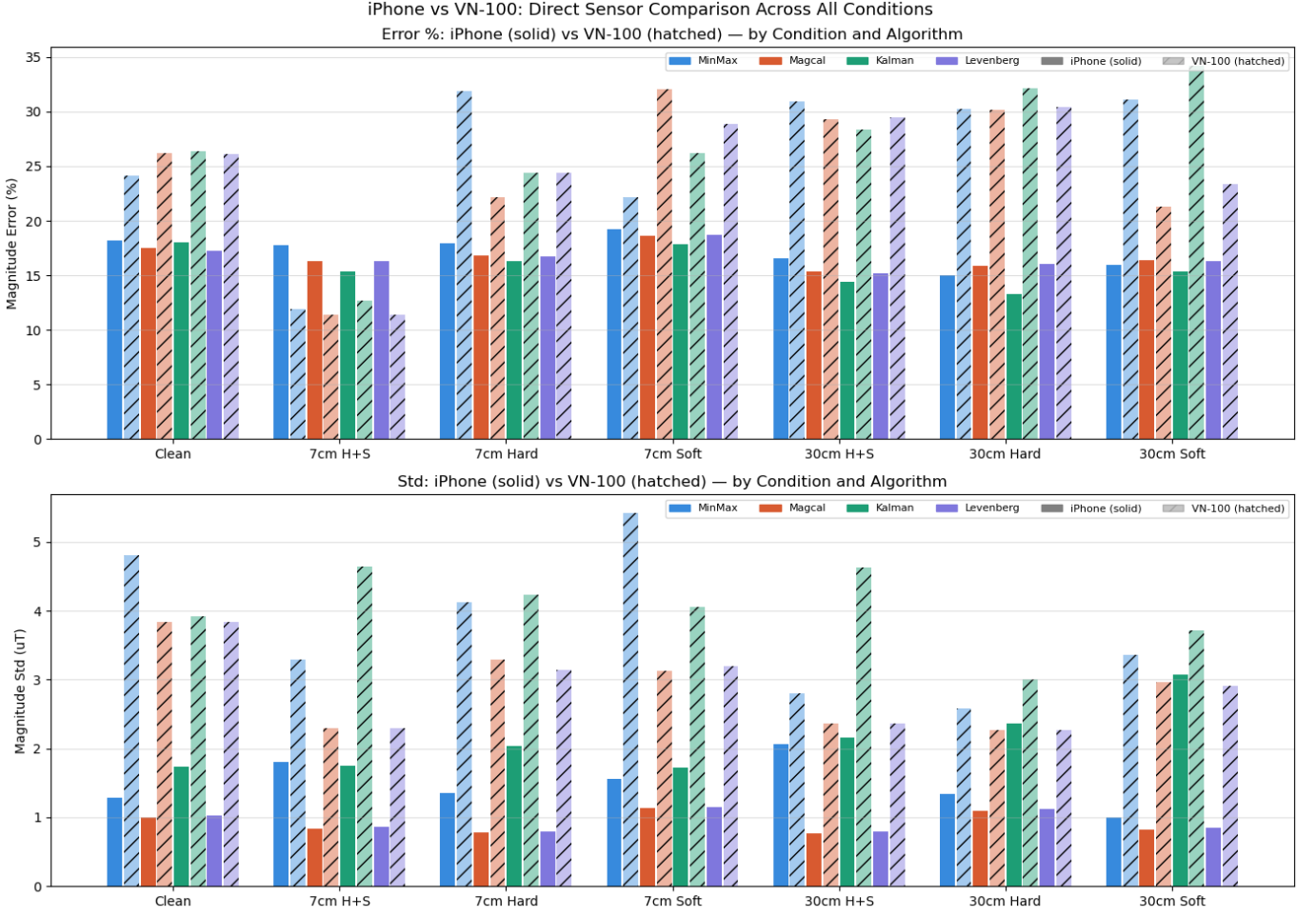


Fig. 4. Sensor Comparison

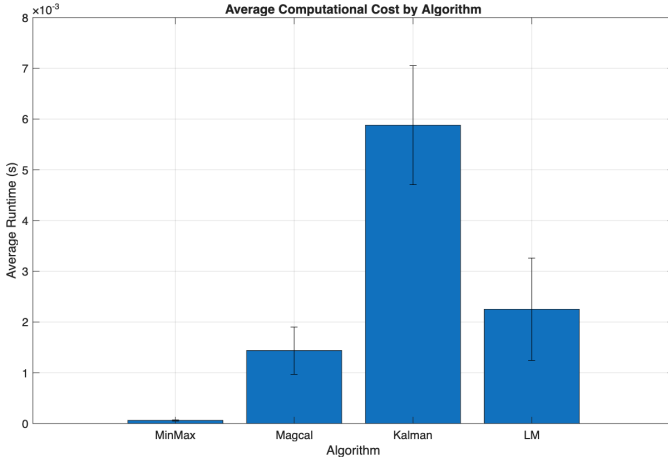


Fig. 5. Algorithm Comparison

algorithms to mitigate. Across most conditions Magcal had the lowest amount of degradation, although Levenberg-Marquardt outperformed Magcal by 0.22% in a hard iron distortion only set up. MinMax showed the greatest sensitivity to distortion intensity, with hard-iron-only and soft-iron-only degradation reaching 2.91% and 3.28% respectively as the distorter moved

from 30cm to 7cm.

The VN-100 had very different results. Under almost all the conditions, the algorithms performed better at seven centimeters than at thirty centimeters. This performance can likely be attributed to the VN-100's internal processing discussed in the Sensor Comparison. The soft-iron-only degradation had the most erratic results with MinMax and the Scalar EKF improving by 8.97% and 8.00% respectively, and Magcal and Levenberg-Marquardt degrading by 10.81% and 5.48% respectively. The Magcal and Levenberg-Marquardt algorithms both use ellipsoid fitting for calibration. This suggests that the Vector Processing Engine degrades the quality of the ellipsoid, reducing the effectiveness of the ellipsoid-based calibration algorithms.

Beyond magnitude error, the standard deviation of the calibrated field magnitude provides insight into calibration consistency across orientations. For the iPhone 11, Magcal and Levenberg-Marquardt achieved the lowest standard deviation at approximately 0.92 μ T and 0.94 μ T respectively, indicating the most consistent correction across all measurement directions. Scalar EKF, despite achieving the lowest mean magnitude error, produced the highest standard deviation at 2.12 μ T, suggesting a tradeoff between accuracy and consistency. This is expected given a sequential estimation.

TABLE I
MAGNITUDE ERROR (%) BY SENSOR, DISTORTION CONDITION, AND DISTANCE

Sensor	Condition	MinMax	Magcal	Kalman	LM
iPhone 11	30cm Hard/Soft	16.56	15.38	14.42	15.19
	7cm Hard/Soft	17.79	16.27	15.38	16.30
	Degradation	+1.23	+0.89	+0.96	+1.11
	30cm Hard Only	15.01	15.89	13.25	16.08
	7cm Hard Only	17.92	16.78	16.33	16.74
	Degradation	+2.91	+0.89	+3.08	+0.66
	30cm Soft Only	15.98	16.43	15.39	16.30
	7cm Soft Only	19.26	18.59	17.84	18.75
	Degradation	+3.28	+2.16	+2.45	+2.45
VN-100	30cm Hard/Soft	30.93	29.31	28.33	29.48
	7cm Hard/Soft	11.92	11.39	12.71	11.43
	Degradation	-19.01	-17.92	-15.62	-18.05
	30cm Hard Only	30.22	30.19	32.15	30.42
	7cm Hard Only	31.89	22.80	24.38	24.42
	Degradation	+1.67	-7.39	-7.77	-6.00
	30cm Soft Only	31.14	21.27	34.20	23.38
	7cm Soft Only	22.17	32.08	26.20	28.86
	Degradation	-8.97	+10.81	-8.00	+5.48

The VN-100 exhibited higher standard deviation across all algorithms, further supporting the hypothesis that the Vector Processing Engine introduces inconsistencies that are difficult for calibration algorithms to correct.

VI. DISCUSSION

A. Limitations and Future Research

Several limitations should be consider when viewing the results of our study. First, our expected field magnitude of Boston (based on the World Magnetic Model) is 52 μT . The exact field magnitude at the recording location was not measured independently, introducing uncertainty in our error calculations.

Second, the data collection relied on manual figure-eight rotations to achieve full sphere coverage. Uneven or incomplete coverage can affect the quality of the ellipsoid fitting algorithm like Magcal and Levenberg-Marquardt. A motorized rotation fixture could ensure consistent and complete coverage.

While the VN-100 vector processing engine has been hypothesized as a source of poor calibration performance, this was not directly verified and collecting data with VPE disabled would answer the question.

It should also be noted that MATLAB's `timeit` function used to calculate the runtimes of each algorithm is not stable. This means the time any given algorithm took to run is not identical across runs. This can be seen in Fig. 5 where the range of run times for a given algorithm is illustrated as well as the average. In the future we can look to investigate memory usage, Floating Point Operations (FLOPs), and how runtime grows with the number of samples collected to get a better understanding of the computational costs associated with each algorithm.

There were also two different locations data was collected in. The majority of the data was collected in Location A, a classroom in Robinson Hall. Two datasets were collected

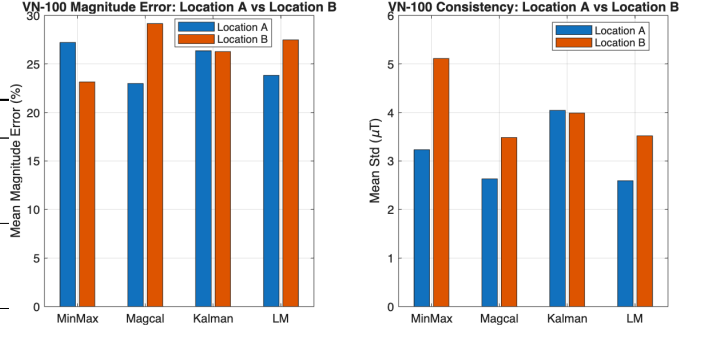


Fig. 6. Location Comparison

at a secondary location, Location B, a common area of an academic building. The VN-100 clean data and VN-100 soft iron distortion data with seven centimeters of separation were collected at Location B. There was not enough time to recollect those two datasets in Location A, so comparison was complicated.

As seen in Fig. 6, data collected in Location B generally produced lower magnitude error but higher standard deviation across all algorithms compared to Location A. This suggests that the magnetic environment at Location B may have been cleaner overall, but the limited variety of distortion conditions collected there, combined with the erratic behavior of the VN-100 under soft iron distortion, resulted in less consistent calibration output.

VII. CONCLUSION

This study compared four magnetometer calibration algorithms across two sensors and fourteen datasets collected under controlled interference conditions. The most significant finding showed the consumer-grade iPhone 11 outperforming the more expensive VN-100 in magnitude error and post calibration consistency. This result is attributed to the VN-100's Vector Processing Engine applying real-time filtering that creates inter-sample dependencies, violating the independence assumption.

Meanwhile among the 4 algorithms, Magcal and Levenburg-Marquardt produced the most consistent calibration output with mean standard deviation of approximately 0.92 μT and 0.94 μT respectively on iPhone data. MinMax was the fastest algorithm due to its use of 6 extreme points, and the Scalar EKF achieved a competitive magnitude error but had the highest standard deviation and worst speed.

Future work should investigate the effect of disabling the VN-100's Vector Processing Engine, implement a motorized rotation fixture for sphere coverage, and evaluate calibration on downstream tasks such as heading estimation error.

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